

# Building Narrative Structures from Knowledge Graphs

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## Overview

- Humans constantly create narratives.
- Narrative defined as a sequence of events, encoded as a graph.

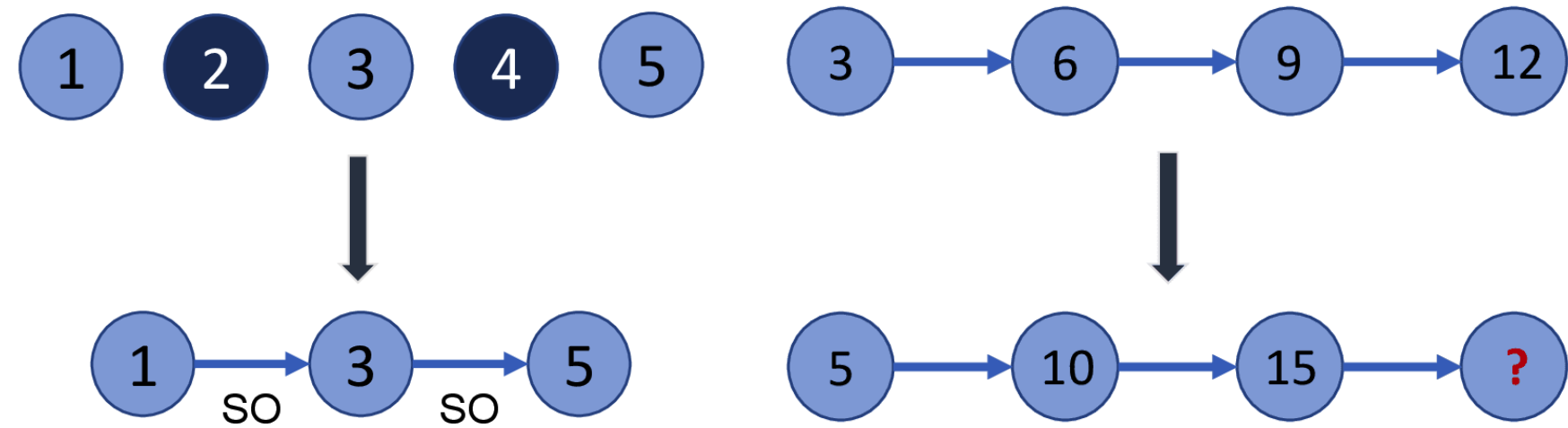


Figure 1: Linking most important events.

Figure 2: Identifying patterns.

## Keywords

Narrative building • Reasoning • Sequence of events • Knowledge Graphs • Temporal, Dynamic, and Event Knowledge Graphs

## Related tasks

Question-Answering • Link prediction • Subgraph extraction • Subgraph prediction • Event recommendation • Temporal graph forecasting

## Research Questions

How to build narrative structures using knowledge graphs as inputs?

- ❶ **RQ-I:** Representation of the Narrative.
- ❷ **RQ-II:** Construction of the Narrative.
- ❸ **RQ-III:** Link Prediction for Narrative Building.
- ❹ **RQ-IV:** Evaluation of the Narrative.

## A Narrative Structure on the French Revolution

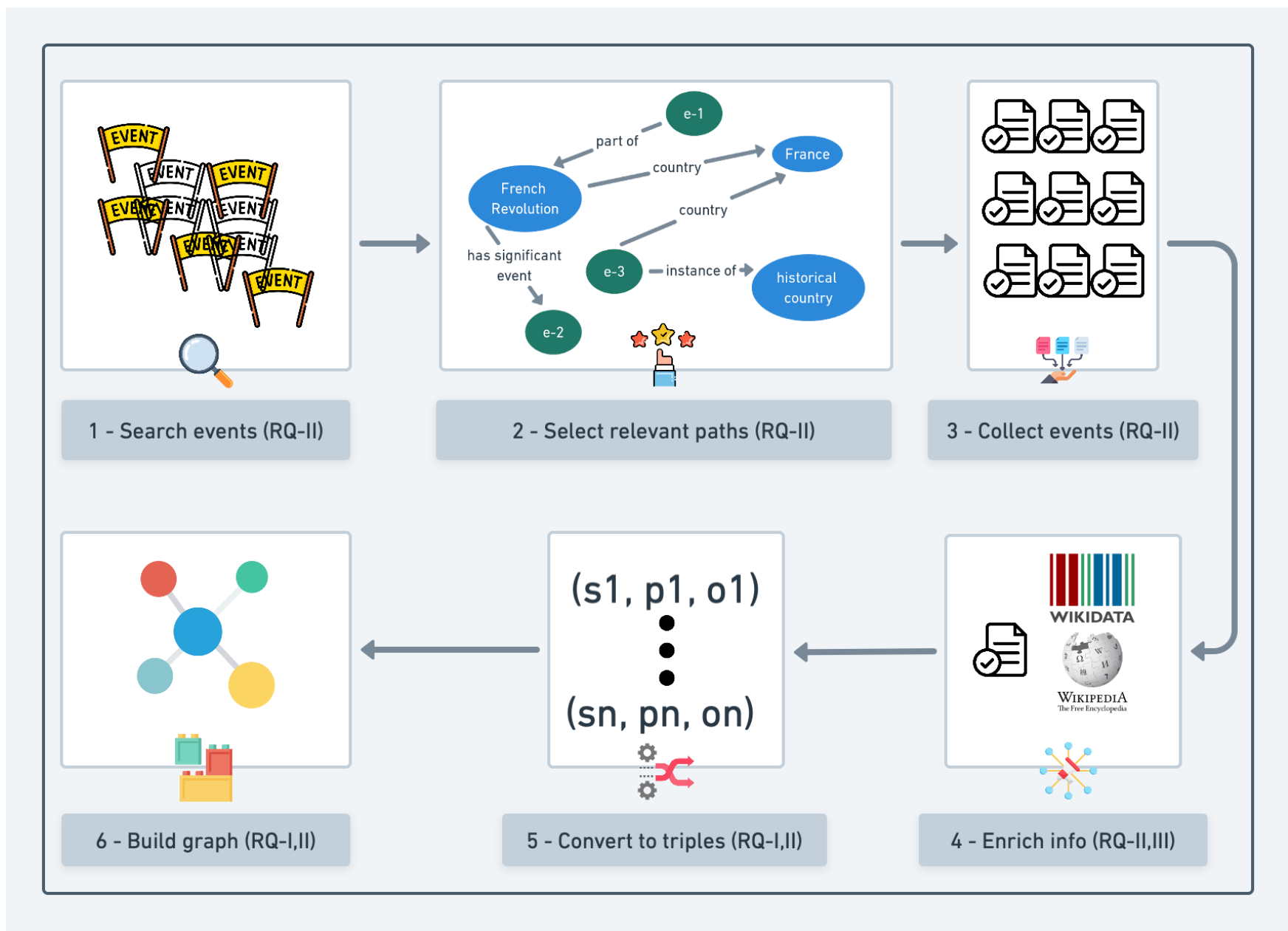


Figure 3: Pipeline of the steps followed to build the final graph.

- Event-level description: who, what, when, where.
- Narrative-level description: temporal and causal links.

| Type        | WD∩WP |       | WD\WP |       | WP\WD |       | Total |       | Not retrieved |       |
|-------------|-------|-------|-------|-------|-------|-------|-------|-------|---------------|-------|
|             | count | perc. | count | perc. | count | perc. | count | perc. | count         | perc. |
| who         | 5     | 10    | 0     | 0     | 15    | 31    | 20    | 42    | 28            | 58    |
| when        | 25    | 52    | 17    | 35    | 0     | 0     | 42    | 87    | 6             | 13    |
| where       | 17    | 35    | 19    | 40    | 4     | 8     | 40    | 83    | 8             | 17    |
| causal link | 1     | 2     | 0     | 0     | 12    | 25    | 13    | 27    | 35            | 73    |
| temp. link  | 21    | 44    | 27    | 56    | 0     | 0     | 48    | 100   | 0             | 0     |

Table 1: Number and percentage of events that contain at least one information from each type. WD = Wikidata, WP = Wikipedia. In total: 48 WD events, 26 WP events.



Figure 4: Visual graph of one event of the French Revolution: 13 Vendémiaire. One can see that the event is a coup d'état with two combatants: the First French Republic and Royalism. One therefore concludes that the event was a conflict between royalists and republicans.

## Conclusion

- First prototype on the French Revolution.
- Automatic data retrieval, manual system design.

## Current & Future Work

- Automate components of the prototype.
- Extracting relevant events from a start node in a knowledge graph.
- Tackle challenges such as temporality, evaluation or granularity.

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